Binary codes closer to the Gilbert-Varshamov bound admit near-linear unique and list decoding

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Editorial note. Complete selected author published code proof faithfully transcribed from CC BY4 formal version, including all local bearing support and AppendixA. Original mathematical expressions, hypotheses and numbered labels preserved; no mathematical repairs or new assumptions supplied. Only display line breaks and ordinary mathematical typography adapt presentation. Reused standard external decoder/base-code results are quoted at the exact original interface. All production and validation solely onYupeng viaSSH.

Allowed standard prerequisites (editorial).

- Bilu-Linial inverse mixing lemma, Hoeffding inequality and explicit ordinary expanders. Exact original cited matrixLemma16 and concentrationFact19 retained; ordinary expander existence is cited as by authors, not recursively reproved
- Guruswami-Indyk base codes and Jeronimo-Srivastava-Tulsiani decoding framework. Original exact Theorems30/35 retained as cited standard inputs; full source-local compatibilityAppendixA and all new sampling/parameter arguments included

2 Preliminaries

For integers a, b, we use $[a, b]$ to denote the set $\{a, \dots, b\}$ and $[n]$ as a shorthand for $[1, n]$. Given a set $S \subseteq [n]$, when the ground set $[n]$ is clear from context, we denote $\rho(S) = |S|/n$, and its ± 1 variant as $\mathrm{bias}(S) = 1 - 2\rho(S)$. For $z \in \mathbb{F}_2^n$, we similarly denote $\mathrm{bias}(z)$ as the bias of its characteristic set, i.e., $\mathbb{E}_{i\in[n]}[(-1)^{z_i}]$. We use boldface letters to denote random variables, except for $\mathbf{1} \in \mathbb{R}^n$, which we use for the all-ones vector. Also, when bounding running time, by writing $g(n) = \exp(f(n))$ we mean that $g(n) \leq 2^{c\cdot f(n)}$ for some universal constant $c > 0$.

Definition5 (discretizable distribution). For $M \in \mathbb{N}$, We say that a distribution $\mathbf{W}$ is M-discretizable if it satisfies either of the following two equivalent properties.

1. For any x in the support of $\mathbf{W}$, $\Pr_{\mathbf{x}\sim\mathbf{W}}[\mathbf{x} = x] = i/M$ for some $i \in \mathbb{N}$.

2. Let U_M be the uniform distribution over $[M]$. Then, there is some function f mapping $[M]$ to the support of $\mathbf{W}$ for which $f(i)$, where $i \sim U_M$, has the same distribution as a sample from $\mathbf{W}$.

We say that $\mathbf{W}$ is computable in (deterministic or probabilistic) time T if f above is computable in time T. In particular, $\mathbf{W}$ is explicit if it is computable in deterministic time $\mathrm{poly}(M)$, and fully explicit if it is computable in deterministic time $\mathrm{poly}(\log M)$.

Definition6 (homogeneous distribution). For $n, t \in \mathbb{N}$, we say that a distribution $\mathbf{W}$ over $[n]^t$ is homogeneous if its restriction to any coordinate is uniform over $[n]$, i.e., if for any $i \in [t]$ and $a \in [n]$ it holds that $\Pr_{\mathbf{w}\sim\mathbf{W}}[\mathbf{w}_i = a] = 1/n$. For any domain $\mathcal{X}$ and two distributions $\mathbf{D}, \mathbf{D'}$ over $\mathcal{X}$ we define the total variation distance of $\mathbf{D}$ and $\mathbf{D'}$ in terms of the optimal test distinguishing the distributions, i.e.,

$$d_{\mathrm{TV}}(\mathbf{D}, \mathbf{D'}) \coloneqq \sup_{T:\mathcal{X}\to[0,1]} \{\mathbb{E}_{\mathbf{x}\sim\mathbf{D}}[T(\mathbf{x})] - \mathbb{E}_{\mathbf{x}\sim\mathbf{D'}}[T(\mathbf{x'})]\}.$$

For a matrix $A \in \mathbb{R}^{n\times n}$, we denote by $\|A\|_{\mathrm{op}}$ its operator norm $\|A\|_{\mathrm{op}} = \max_{x\neq 0} \|Ax\|_2/\|x\|_2$, which is also the maximum of $x^\dagger Ay$ over all norm-1 vectors $x, y \in \mathbb{R}^n$.

Error correcting codes. A binary error correcting code of message length k and blocklength n is a mapping $C : \mathbb{F}_2^k \to \mathbb{F}_2^n$, which we will often identify with its image $\mathrm{Im}(C) \subseteq \mathbb{F}_2^n$. The rate of C is k/n, and its relative distance is $\Delta(C) = \frac{1}{n}\min_{z\neq z'} \Delta(z, z')$ for $z, z' \in C$, and $\Delta(z, z') = |\{i \in [n] : z_i \neq z'_i\}|$ being the Hamming distance. The Hamming ball of (relative) radius β centered at z is the set $B(z, \beta) = \{z' \in \mathbb{F}_2^n : \Delta(z, z')/n \leq \beta\}$. We denote $\mathrm{bias}(C)$ as the maximal bias, in absolute value, of every nonzero $z \in C$. Thus, $\mathrm{bias}(C) \leq \varepsilon$ if the Hamming weight of any nonzero codeword is in $[(1-\varepsilon)/2, (1+\varepsilon)/2]$.

Definition7 (balanced codes). A linear binary error correcting code is ε-balanced if $\mathrm{bias}(C) \leq \varepsilon$. In particular, an ε-balanced code C has distance at least $(1-\varepsilon)/2$.

Unique and list decoding. We say that C is (combinatorially) (β, L) list decodable if for every $z \in \mathbb{F}_2^n$, $|C \cap B(z, \beta)| \leq L$. The Johnson bound tells us that any ε-balanced code is $(1/2 - \sqrt{\varepsilon}, L)$ list decodable for $L = O(1/\varepsilon)$. The algorithmic list decoding problem aims at finding the list $L_{C,\beta}(z) \coloneqq C \cap B(z, \beta)$. When $\beta \leq (1-\varepsilon)/4$ and C is ε-balanced, we know that $L_{C,\beta}$ always contains at most one codeword, which corresponds to the unique decoding problem.

2.1 Parity samplers and direct sum codes

We will be interested in constructing a binary linear code $C \subseteq \mathbb{F}_2^{\bar{n}}$ with small bias by amplifying the (moderate) bias of some base code $C_0 : \mathbb{F}_2^k \to \mathbb{F}_2^n$. One natural way to do so is by XORing t-tuples of C_0 according to some distribution $\mathbf{W} \sim [n]^t$.

Definition8 (direct sum codes). For $t, n, \bar{n} \in \mathbb{N}$, let $\mathbf{W} \sim [n]^t$ be an $\bar{n}$-discretizable distribution equipped with a corresponding mapping function $f_{\mathbf{W}} : [\bar{n}] \to [n]^t$. For $z \in \mathbb{F}_2^n$, we let $\mathrm{dsum}_{\mathbf{W}}(z) \in \mathbb{F}_2^{\bar{n}}$ be such that

$$\mathrm{dsum}_{\mathbf{W}}(z)[\ell] = \sum_{i=1}^{t} z[f_{\mathbf{W}}(\ell)_i]$$

where the addition is taken over $\mathbb{F}_2$. Given a code $C_0 \subseteq \mathbb{F}_2^n$, the direct sum lift of C_0 according to $\mathbf{W}$ is the code $\mathrm{dsum}_{\mathbf{W}}(C_0) = \{\mathrm{dsum}_{\mathbf{W}}(z) : z \in C_0\} \subseteq \mathbb{F}_2^{\bar{n}}$.

Definition9 (parity sampler). For $t, n \in \mathbb{N}$, and $0 \leq \varepsilon < \varepsilon_0 \leq 1$, we say that $\mathbf{W} \sim [n]^t$ is an $(\varepsilon_0, \varepsilon)$ parity sampler if for every $z \in \mathbb{F}_2^n$ with $|\operatorname{bias}(z)| \leq \varepsilon_0$ it holds that $|\operatorname{bias}(\mathrm{dsum}_{\mathbf{W}}(z))| \leq \varepsilon$. Clearly, if $C_0 : \mathbb{F}_2^k \to \mathbb{F}_2^n$ is ε_0-balanced and $\mathbf{W} \sim [n]^t$ is an $\bar{n}$-discretizable $(\varepsilon_0, \varepsilon)$ parity sampler, the lifted code $C = \mathrm{dsum}_{\mathbf{W}}(C_0)$ is ε-balanced with rate $k/\bar{n}$.

3 Expanding 3-Uniform Hypergraphs

Our construction uses a family of expanding d-regular3-uniform hypergraphs.

Definition10 (d-regular3-uniform hypergraph). A3-uniform hypergraph consists of a set of vertices, V, and hyperedges $E \subseteq V^3$. The hypergraph $H = (V, E)$ is d-regular if, for each $v \in V$ and $j \in [3]$, the number of hyperedges $(w_1, w_2, w_3) \in E$ for which $v = w_j$ is d. We will set d carefully in our construction, but for now, it can be thought of as an arbitrary constant. For the remainder of this section, we will use "hypergraph" as shorthand for d-regular3-uniform hypergraph. There are various notions of expansion for hypergraphs and they are not all equivalent. In this work, we consider two such notions.

Definition11 (λ-mixing hypergraph). A d-regular hypergraph $H = (V, E)$ on n vertices is λ-mixing if, for any $S_1, S_2, S_3 \subseteq V$,

$$\left| \frac{E(S_1, S_2, S_3)}{d} - \frac{|S_1| \cdot |S_2| \cdot |S_3|}{n^2} \right| \leq \lambda \sqrt{|S_1| \cdot |S_3|}, \tag{2}$$

where $E(S_1, S_2, S_3)$ is the number of hyperedges $(w_1, w_2, w_3) \in E$ where $w_j \in S_j$ for $j = 1, 2, 3$. Note that the right-hand side of the above definition does not depend on the size of S_2.[1] That parallels the definition below, in which we use $\|y\|_\infty$ instead of $\|y\|_2$.

Definition12 (λ-spectral hypergraph). A d-regular hypergraph $H = (V, E)$ on n vertices is λ-spectral if, for any $x, y, z \in \mathbb{R}^n$,

$$\left| \frac{1}{d} \sum_{(i,j,k) \in E} x_i y_j z_k - \frac{1}{n^2} \sum_{i \in V} x_i \sum_{j \in V} y_j \sum_{k \in V} z_k \right| \leq \lambda \|x\|_2 \|y\|_\infty \|z\|_2. \tag{3}$$

[1] Some works consider the stronger requirement of $\sqrt{|S_{\sigma(1)}| \cdot |S_{\sigma(2)}|}$ instead of $\sqrt{|S_1| \cdot |S_3|}$, for $S_{\sigma(1)}$ and $S_{\sigma(2)}$ being the two smallest sets.

We will only care about cases where $y \in \{\pm 1\}^n$.[2] In those cases, we have $\|y\|_\infty = 1$ and $\|y\|_2 = \sqrt{n}$. It is thus tempting to replace the $\|y\|_\infty$ in the right-hand side of Equation(3) with $\|y\|_2/\sqrt{n}$, as ℓ_2 norms are often easier to work with than ℓ_∞ norms. Unfortunately, no "good" λ-spectral hypergraphs would exist with that modification: Whenever $n \gg d^2$, it is straightforward to bound $\lambda = \Omega(\sqrt{n}/d)$. For our definition, as we shall soon see, it is possible to achieve $\lambda \approx 1/\sqrt{d}$. Similar to standard graphs, spectral expansion implies mixing.

Proposition13 (spectral implies mixing). For any $\lambda > 0$, if H is a λ-spectral hypergraph, it is also a λ-mixing hypergraph.

Proof. For any $S_1, S_2, S_3 \subseteq V$, let $x_i = 1[i \in S_1]$, $y_i = 1[i \in S_2]$, and $z_i = 1[i \in S_3]$. Then, $E(S_1, S_2, S_3) = \sum_{(i,j,k)\in E} x_i y_j z_k$. Equation(2) follows directly from Equation(3). □

By applying the converse to the expander mixing lemma for ordinary graphs[7], we can show that for symmetric hypergraphs, mixing implies spectral expansion with only a minor quantitative gap.

Definition14 (symmetric3-uniform hypergraph). We say a3-uniform hypergraph $H = (V, E)$ is symmetric if, for any edge $e = (v_1, v_2, v_3) \in E$, the edge (v_3, v_2, v_1) is also in E.

Theorem31. There exists an efficient randomized algorithm such that for every$k$ and any$\varepsilon > 0$, outputs with probability$1 - 2^{-\Omega(k)}$ anε-balanced linear code$C : \mathbb{F}_2^k \to \mathbb{F}_2^{\bar{n}}$ of rate

$$\frac{k}{\bar{n}} = 2^{-c_r \log\log(1/\varepsilon)\sqrt{\log(1/\varepsilon)}} \cdot \varepsilon^2$$

for some universal constantc_r, that is encodable in deterministic timeexp(exp($\sqrt{\log(1/\varepsilon)}$))$\cdot k$.[3] More precisely, there exists a randomized preprocessing step that runs in timeexp($\sqrt{\log(1/\varepsilon)}$) $\cdot\, k$ and succeeds with probability$1 - 2^{-\Omega(k)}$. Once it succeeds, it fixes a deterministic mapping$C$ such that for every$x \in \mathbb{F}_2^k$, $C(x)$ can be computed in deterministic in the above$O_\varepsilon(\bar{n})$ time. If, moreover, for a large enough constant$d$ we are given an explicit family ofλ-spectral$d$-regular3-uniform hypergraphs satisfying$\lambda = \text{poly}(\log d)/\sqrt{d}$, there is no need for a preprocessing step, and$C$ is explicit.

Theorem36. Given$k \in \mathbb{N}$ and$\varepsilon > 0$, let$C : \mathbb{F}_2^k \to \mathbb{F}_2^{\bar{n}}$ be theε-balanced, linear-time encodable code guaranteed to us by Theorem31 with high probability. Then,C also admits the following decoding capabilities.

1. C is list decodable up to radius$1/2 - \beta$ for$\beta = 2^{-\frac{1}{2}\sqrt{\log(1/\varepsilon)}}$ by a randomized algorithm that runs in time$c_1(\varepsilon)\cdot\widetilde{O}(k)$ and succeeds with probability$1 - 2^{-\Omega(k)}$.

[2] In fact, for any fixed choice of x and z, the left-hand side of Equation(3) is linear in y. Therefore, it is maximized for some $y \in \{\pm 1\}^n$ and so in general it is sufficient to consider only such y.

[3] Following the footnote on the base code, one can get a randomized encoding in time$\text{poly}(1/\varepsilon) \cdot k$ by using a randomized encoding of the based code.

2. C is uniquely decodable to within distance$(1-\varepsilon)/4$ by a randomized algorithm that runs in time$c_2(\varepsilon)\cdot\widetilde{O}(k)$ and succeeds with probability$1-2^{-\Omega(k)}$.

Above,$c_1(\varepsilon)=\exp(\exp(\exp(\sqrt{\log(1/\varepsilon)})))$ and$c_2(\varepsilon)=\exp(\exp(\sqrt{\log(1/\varepsilon)}))$.

Proposition15 (mixing implies spectral). There exists a universal constant $c_{\text{spec}}>0$ for which the following holds. Let $H=(V=[n],E)$ be any d-regular hypergraph, and $\lambda\le 1/2$. If H is λ-mixing and symmetric, then H is a λ'-spectral for $\lambda'=c_{\text{spec}}\cdot\lambda\log(1/\lambda)$. The proof of Proposition15 is a simple application of a similar result for graphs.

Lemma16 (Lemma3.3 of[7]). There exists a universal constant c_{BL} for which the following hods. For any $n\times n$ real symmetric matrix A and $\lambda\le 1/2$, suppose each row has an ℓ_1 norm of at most1 and for any two vectors $u,v\in\{0,1\}^n$,[4]

$$|u^\dagger Av|\le\lambda\|u\|\|v\|. \tag{4}$$

Then, the spectral radius of A is at most $c_{\text{BL}}\cdot\lambda\log(1/\lambda)$. If G is the (normalized) transition matrix of a d-regular graph, and $A=G-\frac{1}{n}J$ for J being the all-ones matrix, then Lemma16 shows a converse to the expander mixing lemma: Any graph with good mixing is also a good spectral expander. We show that their result can be lifted to3-uniform hypergraphs.

Proof of Proposition15. Fix any $y\in\mathbb{R}^n$ and let $A^{(y)}\in\mathbb{R}^{n\times n}$ be defined as

$$A^{(y)}_{i,k}\coloneqq\frac{1}{2d}\left(\sum_{j\in[n]}1[(i,j,k)\in E]\cdot y_j\right)-\frac{1}{2n^2}\sum_{j\in[n]}y_j.$$

Then, for any $x,z\in\mathbb{R}^n$,

$$2(z^\dagger A^{(y)}x)=\frac{1}{d}\sum_{(i,j,k)\in E}x_iy_jz_k-\frac{1}{n^2}\sum_{i\in V}x_i\sum_{i\in V}y_i\sum_{i\in V}z_j.$$

Therefore, in order to prove that H is an λ'-spectral expander, it is sufficient to show that for all $y\in\mathbb{R}^n$, the operator norm of $A^{(y)}$ is at most $\|y\|_\infty\cdot\lambda'/2$. We observe that:

1. For any fixed $x,z\in\mathbb{R}^n$, the quantity $z^\dagger A^{(y)}x$ is a linear function of y. Thus, we can consider y-s with $\|y\|_\infty=1$ without loss of generality. Furthermore, the y maximizing $z^\dagger A^{(y)}x$ among those with $\|y\|_\infty=1$ will be in $\{\pm1\}^n$. Therefore we assume $y\in\{\pm1\}^n$ also without loss of generality.

2. Since H is symmetric, the operator norm and spectral radius of $A^{(y)}$ are equal, so we instead bound the spectral radius.

We will apply Lemma16 to each $A^{(y)}$. Note that:

[4]Note that Bilu and Linial's Lemma statement only has the weaker requirement that this hold for orthogonal u and v. As a result, they have an additional condition that the diagonal entries of A not be too large, which is not needed for our version of the Lemma.

- $A^{(y)}$ is symmetric, which follows immediately from the fact that H is symmetric.
- The ℓ_1 norm of each row of $A^{(y)}$ is bounded by1:

$$\sum_{k\in[n]}\left|\frac{1}{2d}\left(\sum_{j\in[n]}1[(i,j,k)\in E]\cdot y_j\right)+\frac{1}{2n^2}\sum_{j\in[n]}y_j\right| \leq \frac{1}{2d}\cdot d+\frac{1}{2n^2}\cdot n\leq 1.$$

- Finally, fix some $u,v\in\{0,1\}^n$, and define the sets

  $S_1=\{i\in V: u_i=1\},\quad S_3=\{i\in V: v_i=1\},\quad S_2^+=\{i\in V: y_i=1\},\quad S_2^-=\{i\in V: y_i=-1\}.$

  Then,

$$\begin{aligned}|u^\dagger A^{(y)}v| &= \frac{1}{2}\left|\frac{1}{d}(E(S_1,S_2^+,S_3)-E(S_1,S_2^-,S_3))\right.\\ &\qquad\left.-\frac{|S_1|\cdot|S_3|\cdot(|S_2^+|-|S_2^-|)}{n^2}\right|\\ &\leq \frac{1}{2}\left|\frac{1}{d}E(S_1,S_2^+,S_3)-\frac{|S_1|\cdot|S_3|\cdot|S_2^+|}{n^2}\right|\\ &\quad+\frac{1}{2}\left|\frac{1}{d}E(S_1,S_2^-,S_3)-\frac{|S_1|\cdot|S_3|\cdot|S_2^-|}{n^2}\right|\\ &\leq \lambda\sqrt{|S_1|\cdot|S_3|},\end{aligned}$$

  where the last inequality follows from fact that H is λ-mixing (applied to both expressions).

Thus, $|u^\dagger A^{(y)}v|\leq\lambda\|u\|\|v\|$ and we can apply Lemma16 to obtain $\|A^{(y)}\|_{\mathrm{op}}=c_{\mathrm{BL}}\cdot\lambda\log(1/\lambda)$, implying that H is $(\lambda'=2c_{\mathrm{BL}}\cdot\lambda\log(1/\lambda))$-spectral. □

3.1 Random hypergraphs mix well

In this section, we'll show that given an expanding graph G, its random hypergraph completion is, with high probability, a good expander. Throughout, we say that an undirected regular graph G is a λ-expander if the second largest eigenvalue of its normalized adjacency matrix, in magnitude, is at most λ.

Definition17 (random hypergraph completion of a graph). Let $G=(V=[n],E_G)$ be a d-regular graph. To sample a random hypergraph completion, $\mathbf{H}$ of G, we choose a uniformly random ordering of G's edges, $\{(u_1,v_1),\ldots,(u_{nd},v_{nd})\}=E_G$. Then, we set $\mathbf{H}=(V,E_{\mathbf{H}})$, where

$$E_{\mathbf{H}}\coloneqq\{(u_i,\lceil i/d\rceil,v_i)\mid i\in[nd]\}.$$

Equivalently, for each $(u,v)\in E_G$, we choose an independent and uniform $\mathbf{w}\in V$ and add the hyperedge $(u,\mathbf{w},v)$ to $E_{\mathbf{H}}$, conditioned on the resulting hypergraph $\mathbf{H}$ being d-regular. Next, we prove that the random hypergraph completion mixes well with high probability.

Lemma18. Let $G = (V = [n], E)$ be a d-regular λ-expander and $\mathbf{H}$ be a random hypergraph completion of G. With probability at least $1 - 2^{-n}$, $\mathbf{H}$ is a λ'-mixing hypergraph for $\lambda' = 2\lambda + 2/\sqrt{d}$. In order to prove Lemma18, we'll use Hoeffding's inequality for sampling without replacements.

Fact19 (Hoeffding's inequality,[18]). For any integers $a, k \leq m$, suppose there are m items, of which k of them are marked. Let $\mathbf{x}$ be a random variable indicating the number of marked items when a of the m items are sampled uniformly and independently without replacement. Then, for any $t \geq 0$,

$$\Pr[|\mathbf{x} - (k/m) \cdot a| \geq t] \leq 2\exp(-2t^2/a).$$

Proof of Lemma18. Fix arbitrary $S_1, S_2, S_3 \subseteq V$. We will show that with probability at least $1 - 2^{-4n}$ it holds that

$$\left| \frac{E_{\mathbf{H}}(S_1, S_2, S_3)}{d} - \frac{|S_1| \cdot |S_2| \cdot |S_3|}{n^2} \right| \leq \left(\frac{2}{\sqrt{d}} + 2\lambda \right) \sqrt{|S_1| \cdot |S_3|} \tag{5}$$

where $E_{\mathbf{H}}(S_1, S_2, S_3)$ is the number of edges (v_1, v_2, v_3) in $\mathbf{H}$ where $v_j \in S_j$ for each $j \in [3]$. The desired result then follows from a union bound over the $(2^n)^3 = 2^{3n}$ choices for S_1, S_2, S_3. Let $E_G(S_1, S_3)$ be the number of edges, (u, v), of G, such that $u \in S_1$ and $v \in S_3$. By the expander mixing lemma applied to G, we have that

$$\left| \frac{E_G(S_1, S_3)}{d} - \frac{|S_1||S_3|}{n} \right| \leq \lambda \sqrt{|S_1||S_3|}.$$

Let us define $\mu \coloneqq |S_1||S_3|/n$ and $\Delta \coloneqq \lambda\sqrt{|S_1||S_3|}$. We consider two cases.

1. In the first case, $\mu \leq \Delta$. Here, we use the simple bound

   $$0 \leq \frac{E_{\mathbf{H}}(S_1, S_2, S_3)}{d} \leq \frac{E_G(S_1, S_3)}{d} \leq 2\Delta$$

   that holds with probability1. This, along with the fact $|S_1| \cdot |S_2| \cdot |S_3|/n^2 \leq \mu \leq \Delta$ implies that Equation(5) always holds.

2. In the second case, $\mu > \Delta$. Here, we will apply Fact19. G has a total of nd edges, of which $d \cdot |S_2|$ are matched to a vertex in S_2. $E_{\mathbf{H}}(S_1, S_2, S_3)$ samples $E_G(S_1, S_3)$ of those nd edges (without replacement) and counts how many were among the $d \cdot |S_2|$ assigned to S_2. Hence, by Hoeffding's inequality, we have for any $t \geq 0$,

   $$\Pr\left[\left| E_{\mathbf{H}}(S_1, S_2, S_3) - \frac{d \cdot |S_2|}{nd} \cdot E_G(S_1, S_3) \right| \geq t \right] \leq 2\exp\left(\frac{-2t^2}{E_G(S_1, S_3)} \right).$$

   Then, setting $t = 2\sqrt{d|S_1| \cdot |S_3|}$ and using the fact that $E_G(S_1, S_3) \leq d(\mu + \Delta) \leq 2d\mu$,

   $$\Pr\left[\left| E_{\mathbf{H}}(S_1, S_2, S_3) - \frac{|S_2|}{n} \cdot E_G(S_1, S_3) \right| \geq 2\sqrt{d|S_1| \cdot |S_3|} \right]$$
   $$\leq 2\exp\left(\frac{-8d|S_1| \cdot |S_3|}{2d \cdot (|S_1||S_3|/n)} \right) = 2\exp(-4.5n) \leq 2^{-4n}.$$

As the above shows, $E_{\mathbf{H}}(S_1, S_2, S_3)$ is within $\pm t$ of its expectation with probability at least 2^{-4n}. When that occurs, we have that

$$\begin{aligned}
&\left|\frac{E_{\mathbf{H}}(S_1,S_2,S_3)}{d} - \frac{|S_1|\cdot|S_2|\cdot|S_3|}{n^2}\right| \\
&= \frac{1}{d}\left|E_{\mathbf{H}}(S_1,S_2,S_3) - \frac{|S_2|d\mu}{n}\right| \\
&\le \frac{t}{d} + \frac{|S_2|}{n}\left|\frac{E_G(S_1,S_3)}{d} - \mu\right| \\
&\le \frac{t}{d} + \frac{|S_2|}{n}\cdot\Delta \qquad \text{(expander mixing lemma)} \\
&\le 2\sqrt{\frac{|S_1|\cdot|S_3|}{d}} + \lambda\sqrt{|S_1|\cdot|S_3|}. \qquad (|S_2|/n \le 1)
\end{aligned}$$

Hence, Equation(5) holds with probability at least $1 - 2^{-4n}$ in both cases, so we can union bound over the 2^{3n} choices for S_1, S_2, S_3. □

For our purposes, we will want the hypergraph to be symmetric. It is quite easy to "symmetrize" any hypergraph at only a modest cost to the degree.

Proposition20. For any $n, d \in \mathbb{N}$, there is an algorithm running in time $O(nd)$ that takes as input any d-regular λ-mixing hypergraph $H = (V = [n], E_H)$ and outputs a $2d$-regular λ-mixing hypergraph $H' = (V = [n], E_{H'})$ over the same vertices.

Proof. For every edge $(u, v, w) \in E_H$ we include both (u, v, w) and the reverse edge (w, v, u) in $E_{H'}$.[5] Clearly, this results in the degree of H' being $2d$. Then, for any $S_1, S_2, S_3 \subseteq V$,

$$\begin{aligned}
&\left|\frac{E_{H'}(S_1,S_2,S_3)}{2d} - \frac{|S_1|\cdot|S_2|\cdot|S_3|}{n^2}\right| \\
&= \left|\frac{E_H(S_1,S_2,S_3) + E_H(S_3,S_2,S_1)}{2d} - \frac{|S_1|\cdot|S_2|\cdot|S_3|}{n^2}\right| \\
&\le \frac{1}{2}\left|\frac{E_H(S_1,S_2,S_3)}{d} - \frac{|S_1|\cdot|S_2|\cdot|S_3|}{n^2}\right| \\
&\quad + \frac{1}{2}\left|\frac{E_H(S_3,S_2,S_1)}{d} - \frac{|S_1|\cdot|S_2|\cdot|S_3|}{n^2}\right| \\
&\le \lambda\sqrt{|S_1|\cdot|S_3|}.
\end{aligned}$$

Therefore, H' is λ-mixing, as desired. □

We have explicit (and even fully explicit) constructions of Ramanujan graphs, i.e., λ-expanders for $\lambda \le 2\sqrt{d-1}/d$, albeit with some restrictions on d[26,28]. By manipulating Ramanujan graphs, Alon gave a construction of d-regular λ-expanders over n vertices for any d and n while suffering only a tiny loss in λ (see[1], and also[29,13] for weaker constructions). In particular, there exist explicit expanders with $\lambda = O(1/\sqrt{d})$ for all n-s. We thus get the following corollary.

[5] Note that, we do this even if it results in duplicated hyperedges: If (u, v, w) and (w, v, u) are both in E_H, then there will be two copies of (u, v, w) and two copies of (w, v, u) in $E_{H'}$.

Corollary21. There exists a probabilistic algorithm such that for any integer n and even integer $6 \leq d \leq n$, runs in time poly(n) and with probability at least $1 - 2^{-n}$ outputs a3-uniform symmetric d-regular hypergraph that is $\lambda = c_{\text{rand}}/\sqrt{d}$-mixing, where $c_{\text{rand}} \geq 2$ is some universal constant. We will refer to the above probabilistic construction as our preprocessing step.

4 From Hypergraphs to Parity Samplers

Fix some $n, t, d \in \mathbb{N}$ and a d-regular3-uniform hypergraph $H = (V, E_H)$ over n vertices. We will construct a parity sampler $\mathbf{W}_{H,t} \sim [n]^t$ from H and show that when H is a good spectral expander, $\mathbf{W}$ is a good parity sampler.

The construction. For each $v \in V$, there are d edges $(v_1, v_2, v_3) \in E_H$ with $v_1 = v$. For any $i \in [d]$, let $e_H(v, i)$ be the ith such edge (i.e., $e_H(v, i)_1 = v$). Without loss of generality, we assume the vertices are each labeled with a unique integer between1 and$n$ (i.e $V = [n]$). To sample $\mathbf{w} \sim \mathbf{W}_{H,t}$, independently sample a starting vertex $\mathbf{v}_0 \in [n]$ and edge labels $\mathbf{i}_1, \ldots, \mathbf{i}_t \sim [d]$ uniformly. $\mathbf{w}$ will be a deterministic function of $\mathbf{v}_0$ and $\mathbf{i}_1, \ldots, \mathbf{i}_t$ computed as follows. For each $j = 1, \ldots, t$,

1. Let $\mathbf{e}_j = e_H(\mathbf{v}_{j-1}, \mathbf{i}_j)$.
2. Let $\mathbf{v}_j = (\mathbf{e}_j)_3$.
3. Let $\mathbf{w}_j = (\mathbf{e}_j)_2$.

The sample is then $\mathbf{w} = (\mathbf{w}_1, \ldots, \mathbf{w}_t)$. We present two simple claims about $\mathbf{W}_{H,t}$.

Claim23. For any $t \in \mathbb{N}$ and a d-regular hypergraph H over n vertices, $\mathbf{W}_{H,t}$ is (nd^t)-discretizable.

Proof. The sample $\mathbf{w} \sim \mathbf{W}_{H,t}$ is a deterministic function of $\mathbf{v}_0$ and $\mathbf{i}_1, \ldots, \mathbf{i}_t$, and those variables are set to a uniform choice out of nd^t possibilities. The claim then follows from Definition5. □

Claim24. For any $t \in \mathbb{N}$, d-regular hypergraph H, and $j \in [t]$, $\mathbf{e}_j$ is uniform over all nd edges of H. As a result, $\mathbf{W}_{H,t}$ is homogeneous.

Proof. We will first prove that each $\mathbf{e}_j$ is uniform over E_H by induction on$j$. $\mathbf{e}_1$ is uniform over the nd edges in E_H as it is sampled by independently choosing a starting vertex $\mathbf{v}_0 \sim V$ uniformly and then uniformly choosing one of its d-neighbors. Furthermore, for any $j \in [t-1]$, if $\mathbf{e}_j$ is uniform, then $\mathbf{v}_j$ is also uniform. Hence, $\mathbf{e}_{j+1}$ is sampled by selecting a uniform vertex and then (independently) one of its d-neighbors, so $\mathbf{e}_{j+1}$ is also uniform over E_H. Next, we show $\mathbf{W}_{H,t}$ is homogeneous. Fix any $a \in [n]$ and $j \in [t]$. As $\mathbf{w}_j = a$ if and only if $\mathbf{e}_j$ is one of the d-edges in E_H whose second vertex is$a$ and $\mathbf{e}_j$ is uniform over the nd edges in E_H, $\Pr[\mathbf{w}_j = a] = d/(nd) = 1/n$. □

Finally, we show that whenever H is a good expander, $\mathbf{W}_{H,t}$ is a good parity sampler.

Theorem25. For any $n, d, t \in \mathbb{N}$, $\lambda, \varepsilon_0 > 0$, and H a λ-spectral d-regular3-uniform hypergraph on$n$ vertices, $\mathbf{W}_{H,t}$ is an $(\varepsilon_0, \varepsilon \coloneqq (\varepsilon_0 + \lambda)^t)$-parity sampler. As an immediate corollary of Theorem25 and Proposition15:

Corollary26. There exists an absolute constant $c_{\text{spec}} > 0$ for which the following holds. For any $n, d, t \in \mathbb{N}$, $\lambda, \varepsilon_0 > 0$, and H a λ-mixing symmetric d-regular3-uniform hypergraph on$n$ vertices, $\mathbf{W}_{H,t}$ is an $(\varepsilon_0, \varepsilon \coloneqq (\varepsilon_0 + c_{\text{spec}} \cdot \lambda \log(1/\lambda))^t)$-parity sampler. Throughout the remainder of this section, we will use the shorthand $\mathbf{W} \coloneqq \mathbf{W}_{H,t}$. For proving Theorem25, it will be convenient to consider $\sigma \in \{\pm 1\}^n$ rather than $z \in \mathbb{F}_2^n$ (as in Definition9) using the mapping $\sigma_i = (-1)^{z_i}$. Equivalent to Definition9, $\mathbf{W}$ is an $(\varepsilon_0, \varepsilon)$ parity sampler if $|\operatorname{bias}_{\mathbf{W}}(\sigma)| \leq \varepsilon$ for all $\sigma \in \{\pm 1\}^n$ satisfying $|\operatorname{bias}(\sigma)| = |\mathbb{E}_{i \sim [n]}[\sigma_i]| \leq \varepsilon_0$, where

$$\operatorname{bias}_{\mathbf{W}}(\sigma) \coloneqq \mathbb{E}_{\mathbf{w} \sim \mathbf{W}} \left[\prod_{j=1}^{t} \sigma_{\mathbf{w}_j} \right].$$

Similarly to[34], we express that bias algebraically.

Lemma27. Let $A^{(\sigma)} \in \mathbb{R}^{n \times n}$ be defined as

$$A^{(\sigma)}_{i,k} \coloneqq \frac{1}{d} \sum_{(i',j',k') \in E_H} \sigma_{j'} \cdot 1[i' = i \wedge k' = k].$$

Then, $\operatorname{bias}_{\mathbf{W}}(\sigma) = \frac{1}{n} \mathbf{1}^\dagger (A^{(\sigma)})^t \mathbf{1}$.

Proof. Let $\mathbf{v}_0, \ldots, \mathbf{v}_t$ and $\mathbf{w}_1, \ldots, \mathbf{w}_t$ be the random variables defined in the construction of $\mathbf{W}$. We claim that for each $j \in \{0, 1, \ldots, t\}$ and$v \in [n]$, that

$$\mathbb{E}\left[1[\mathbf{v}_j = v] \prod_{k=1}^{j} \sigma_{\mathbf{w}_k} \right] = \frac{1}{n} \left(\mathbf{1}^\dagger (A^{(\sigma)})^j \right)_v \tag{6}$$

By induction on$j$. Clearly, for$j = 0$, Equation(6) holds as both sides are equal to$1/n$ for any$v \in [n]$. For$j \geq 1$,

$$\begin{aligned}
&\mathbb{E}\left[1[\mathbf{v}_j = v] \prod_{k=1}^{j} \sigma_{\mathbf{w}_k} \right] \\
&= \sum_{v' \in [n]} \mathbb{E}\left[1[\mathbf{v}_j = v \wedge \mathbf{v}_{j-1} = v'] \prod_{k=1}^{j} \sigma_{\mathbf{w}_k} \right] \\
&= \sum_{v' \in [n]} \mathbb{E}\left[1[\mathbf{v}_{j-1} = v'] \prod_{k=1}^{j-1} \sigma_{\mathbf{w}_k} \right] \mathbb{E}[1[\mathbf{v}_j = v] \sigma_{\mathbf{w}_j} \mid \mathbf{v}_{j-1} = v'] \\
&= \sum_{v' \in [n]} \frac{1}{n} \left(\mathbf{1}^\dagger (A^{(\sigma)})^{j-1} \right)_{v'} \mathbb{E}[1[\mathbf{v}_j = v] \sigma_{\mathbf{w}_j} \mid \mathbf{v}_{j-1} = v'] \\
&= \frac{1}{n} \sum_{(a,b,c) \in E_H} \left(\mathbf{1}^\dagger (A^{(\sigma)})^{j-1} \right)_a \cdot \frac{1}{d} \cdot \sigma_b \cdot 1[v = c] \\
&= \frac{1}{n} \left(\mathbf{1}^\dagger (A^{(\sigma)})^j \right)_v,
\end{aligned}$$

where the third equality is the inductive hypothesis. Then,

$$\begin{aligned}\operatorname{bias}_{\mathbf{W}}(\sigma) &= \mathbb{E}_{\mathbf{w}\sim\mathbf{W}}\left[\prod_{j=1}^{t}\sigma_j\right] \\ &= \sum_{v\in[n]}\mathbb{E}\left[1[\mathbf{v}_j = v]\prod_{k=1}^{j}\sigma_{\mathbf{w}_k}\right] = \frac{1}{n}\mathbf{1}^\dagger(A^{(\sigma)})^t\mathbf{1}.\end{aligned}$$

□

Next, we use the fact that$H$ is aλ-spectral expander to reason about$A^{(\sigma)}$.

Proposition28. Let$J_n \in \mathbb{R}^{n\times n}$ be the matrix in which every element is$1/n$. For any$\sigma \in \{\pm 1\}^n$,

$$\|A^{(\sigma)} - \operatorname{bias}(\sigma)J_n\|_{\mathrm{op}} \le \lambda.$$

Proof. Fix any$x, z \in \mathbb{R}^n$. Then,

$$\begin{aligned}z^\dagger(A^{(\sigma)} - \operatorname{bias}(\sigma)J_n)x &= \frac{1}{d}\sum_{(i,j,k)\in E_H} x_k\sigma_j z_i - \frac{\operatorname{bias}(\sigma)}{n}\sum_{i\in[n]}x_i\sum_{j\in[n]}z_j \\ &= \frac{1}{d}\sum_{(i,j,k)\in E_H} x_i\sigma_j z_k - \frac{1}{n^2}\sum_{k\in[n]}\sigma_k\sum_{i\in[n]}x_i\sum_{j\in[n]}z_j \\ &\le \lambda\|x\|_2\|z\|_2. \qquad \text{(Definition12)}\end{aligned}$$

□

As an immediate consequence, we have:

Corollary29. For any$\sigma \in \{\pm 1\}^n$, $\|A^{(\sigma)}\|_{\mathrm{op}} \le |\operatorname{bias}(\sigma)| + \lambda$.

Proof. As$\|J_n\|_{\mathrm{op}} = 1$, the desired result follows from the reversed triangle inequality applied to the operator norm. □

Finally, we prove Theorem25.

Proof of Theorem25. For any$\sigma \in \{\pm 1\}^n$ satisfying$|\operatorname{bias}(\sigma)| \le \varepsilon_0$, we have:

$$\begin{aligned}|\operatorname{bias}_{\mathbf{W}}(\sigma)| &= \frac{\mathbf{1}^\dagger}{\sqrt{n}}(A^{(\sigma)})^t\frac{\mathbf{1}}{\sqrt{n}} \qquad \text{(Lemma27)} \\ &\le \|(A^{(\sigma)})^t\|_{\mathrm{op}} \le \|A^{(\sigma)}\|_{\mathrm{op}}^t \\ &\le (|\operatorname{bias}(\sigma)| + \lambda)^t \le (\varepsilon_0 + \lambda)^t. \qquad \text{(Corollary29)}\end{aligned}$$

□

5 Codes Closer to the GV Bound

5.1 The construction

We use our parity sampler to amplify the distance of a given base code via direct sum lifting. More formally, given$k \in \mathbb{N}$ and$\varepsilon > 0$, let$\varepsilon_0 = \varepsilon_0(\varepsilon)$ soon to be determined, and let

- $C_0 : \mathbb{F}_2^k \to \mathbb{F}_2^n$ be anε_0-balanced code, and,
- $\mathbf{W} \sim [n]^t$ be an$(\varepsilon_0, \varepsilon)$-parity sampler which is$M$-discretizable.

Denote$\bar{n} = |\operatorname{Supp}(\mathbf{W})| \leq M$. Thus, the lifted code$C = \operatorname{dsum}_{\mathbf{W}}(C_0) \subseteq \mathbb{F}_2^{\bar{n}}$ is clearlyε-balanced. For the base code, we use the codes by Guruswami and Indyk, that admit$O_{\varepsilon_0}(n)$-time encoding and decoding.

Theorem30 ([15]). For every integer$n$ and any$\varepsilon_0 > 0$ there exists anε_0-balanced code$C_0 \subseteq \mathbb{F}_2^n$ of rate$\Omega(\varepsilon_0^3)$. Furthermore,C_0 is encodable in time$\exp(\operatorname{poly}(1/\varepsilon_0))n$ and decodable from$1/4 - \varepsilon_0$ fraction of errors in time$\exp(\operatorname{poly}(1/\varepsilon_0))n$.[6]

Setting parameters. Given$\varepsilon > 0$, we henceforth set:

1. The initial bias ofC_0 to$\varepsilon_0 = \frac{1}{2} \cdot 2^{-\sqrt{\log(1/\varepsilon)}}$.
2. The number of steps over$H$ to$t = \lceil \sqrt{\log(1/\varepsilon)} \rceil$.
3. The degree$d$ of$H$ to be the smallest even integer for which

$$c_{\text{spec}} \cdot c_{\text{rand}} \cdot \frac{1}{\sqrt{d}} \log\left(\frac{\sqrt{d}}{c_{\text{rand}}}\right) \leq \varepsilon_0.$$

   This is chosen so that Corollary21 gives, with high probability, a symmetric hypergraph$H$ such that, by Corollary26,$\mathbf{W}_{H,t}$ is a$(\varepsilon_0, (2\varepsilon_0)^t)$-parity sampler.

Using the above parameters in Theorem25, together with the random hypergraph of Corollary21. we get the following (randomized) error correcting code$C$ [Theorem31, stated above].

Proof of Theorem31. Given$k \in \mathbb{N}$ and$\varepsilon > 0$, we setε_0, t, and$d$ as above. Theorem30 gives us a code$C_0 : \mathbb{F}_2^k \to \mathbb{F}_2^n$ which isε_0-balanced, for$n = O(k/\varepsilon_0^3)$. By Corollary21 we can output a$\lambda = c_{\text{rand}}/\sqrt{d}$-mixing$H$ with probability at least$1 - 2^{-n}$ in polynomial time. This is our preprocessing step. By Corollary26 and our choice of parameters, indeed

$$\left(\varepsilon_0 + c_{\text{spec}} \cdot \lambda \log \frac{1}{\lambda}\right)^t \leq (2\varepsilon_0)^t \leq \varepsilon,$$

so$\mathbf{W} = \mathbf{W}_{H,t}$ appropriately amplifies the distance. Moreover,$\mathbf{W}$ is$M$-discretizable for

$$\begin{aligned} M = nd^t &= O\left(\frac{k}{\varepsilon_0^3}\right) \cdot d^{\lceil \sqrt{\log(1/\varepsilon)} \rceil} \\ &= k \cdot 2^{(3+\log d)\sqrt{\log(1/\varepsilon)} + \log d + O(1)}. \end{aligned}$$

[6] One can get a better randomized encoding time of$\operatorname{poly}(1/\varepsilon_0)$.

We proceed to boundinglog d in terms ofε. Recalling how we set$d$, we see that$\frac{c}{\sqrt{d}} \log d = 2^{-\sqrt{\log(1/\varepsilon)}}$, where$c$ is some universal positive constant. From this we can infer that

$$d \leq c^2 2^{2\sqrt{\log(1/\varepsilon)}} \cdot \log^3 \left(c \cdot 2^{\sqrt{\log(1/\varepsilon)}} \right), \tag{7}$$

solog $d = 2\sqrt{\log(1/\varepsilon)} + O(\log\log(1/\varepsilon))$. (We are assuming that$\varepsilon$ is smaller than some small constant, which we can without loss of generality.) Hence,

$$M = \frac{k}{\varepsilon^2} \cdot 2^{O(\sqrt{\log(1/\varepsilon)} \cdot \log\log(1/\varepsilon))}.$$

Since$\mathbf{W}$ is$M$-discretizable,$C = \mathrm{dsum}_{\mathbf{W}}(C_0)$ has block length$\bar{n} \leq M$, as required. Drawing$H$ at random, by Corollary21 takes

$$O(dn) = \widetilde{O}\left(\frac{1}{\varepsilon_0^3}\right) \cdot k = \exp(\sqrt{\log(1/\varepsilon)}) \cdot k$$

time. Given the hypergraph$H$, the encoding amounts to computing$C_0(x)$ and taking parities of its coordinates according to$\mathbf{W}$. This takesexp(poly$(1/\varepsilon_0)) \cdot k + M \cdot t$ time, following Theorem30. For the "moreover" part, one can verify that we can tolerate apolylog(d) multiplicative factor inλ with no substantial loss in parameters. □

5.2 τ-sampling

Toward establishing efficient decoding, we will need the notion ofτ-sampling.

Definition32 (τ-sampling). We say that a distribution$\mathbf{W}$ over$[n]^t$ isτ-sampling if for any$i \in [t-1]$, $S \subseteq [n]$, and$X \subseteq [n]^i$,

$$\mathrm{Cov}_{\mathbf{w} \sim \mathbf{W}}[1[\mathbf{w}_{i+1} \in S], 1[(\mathbf{w}_1, \ldots, \mathbf{w}_i) \in X]] \leq \tau.$$

To highlight the fact that it is indeed a (strong) sampling property, assume for simplicity that$\mathbf{W}$ is homogeneous. Fix any$i \in [t-1]$, $S \subseteq [n]$, and$X \subseteq [n]^i$. The property ofτ-sampling thus tells us we can use$\mathbf{W}$ to sample$S$ starting from any prefix. Namely, that

$$\left| \Pr_{\mathbf{w} \in \mathbf{W}}[\mathbf{w}_{i+1} \in S \mid (\mathbf{w}_1, \ldots, \mathbf{w}_i) \in X] - \rho(S) \right| \leq \frac{\tau}{\rho(X)}.$$

Note that the$t = 2$ case corresponds to the setting of the expander mixing lemma. Aτ-sampling distribution$\mathbf{W}$ satisfies the property of the "Splittable Mixing Lemma" of Jeronimo, Srivastava, and Tulsiani [20,Lemma4.6] for the family of "± 1 cut functions". This fact is crucial us, and for completeness we establish this in AppendixA. Given aτ-sampling homogeneous$\mathbf{W} \sim [n]^t$, a standard hybrid argument shows the following.

Lemma33. Let$\mathbf{W} \sim [n]^t$ be homogeneous andτ-sampling, and let$z \in \mathbb{F}_2^n$ be arbitrary. Then,

$$\left| \mathbb{E}_{\mathbf{w} \in \mathbf{W}}[(-1)^{z_{\mathbf{w}_1} + \cdots + z_{\mathbf{w}_t}}] - \mathrm{bias}(z)^t \right| \leq 2(t-1)\tau.$$

Thus, high-order mixing in particular implies that$\mathbf{W}$ is an$(\varepsilon_0, \varepsilon = \varepsilon_0^t + (t-1)\tau)$ parity sampler for allε_0. However, for us, and also in[20], $t \cdot \tau$ is too large so we prove the parity sampling property separately. We conclude our discussion aboutτ-sampling by showing that our$\mathbf{W}$ is indeedτ-sampling.

Lemma34. For any$n, d, t \in \mathbb{N}$ and$\lambda > 0$, if$H = (V = [n], E)$ is aλ-spectral$d$-regular3-uniform hypergraph, then$\mathbf{W}_{H,t}$ is$\tau = \lambda/4$-sampling.

Proof. As in Definition32, fix any$i \in [t-1]$, $S \subseteq [n]$ and$X \subseteq [n]^i$. Let$S_i(\mathbf{w})$ be the indicator that$\mathbf{w}_{i+1} \in S$, and$X_i(\mathbf{w})$ the indicator that$(\mathbf{w}_1, \ldots, \mathbf{w}_t) \in X$. Using the identity$\mathrm{Cov}[1-2a, 1-2b] = 4\,\mathrm{Cov}[a, b]$, it is sufficient to prove that

$$\mathrm{Cov}_{\mathbf{w} \sim \mathbf{W}_{H,t}}[(-1)^{S_i(\mathbf{w})}, (-1)^{X_i(\mathbf{w})}] \leq \lambda.$$

Let$\mathbf{v}_0, \ldots, \mathbf{v}_t$, $\mathbf{w}_1, \ldots, \mathbf{w}_t$, and$\mathbf{e}_1, \ldots, \mathbf{e}_t$ be the random variables defined in the construction of$\mathbf{W}_{H,t}$. Furthermore, let$x, y, z \in \mathbb{R}^n$ be the vectors defined, for each$j \in [n]$, by

$$x_j \coloneqq \mathbb{E}[(-1)^{X_i(\mathbf{w})} \mid \mathbf{v}_i = j], \qquad y_j \coloneqq (-1)^{1[j \in S]}, \qquad z_j \coloneqq 1/n.$$

Note that$\|z\|_2 = 1/\sqrt{n}$, $\|y\|_\infty = 1$, and$\|x\|_2 \leq \sqrt{n}$ (which follows from$\|x\|_\infty \leq 1$). Our goal in this proof will be to show that the following equation holds:

$$\mathrm{Cov}_{\mathbf{w}}[(-1)^{S_i(\mathbf{w})}, (-1)^{X_i(\mathbf{w})}] = \frac{1}{d} \sum_{(a,b,c) \in E} x_a y_b z_c - \frac{1}{n^2} \sum_{a \in [n]} x_a \sum_{a \in [n]} y_a \sum_{a \in [n]} z_a. \quad (8)$$

Once we do, the desired result follows from Definition12. In order to compute the covariance, we first expand:

$$\begin{aligned}
&\mathbb{E}_{\mathbf{w}}[(-1)^{S_i(\mathbf{w})} \cdot (-1)^{X_i(\mathbf{w})}] \\
&= \frac{1}{nd} \sum_{e \in E} \mathbb{E}_{\mathbf{w}}[(-1)^{S_i(\mathbf{w})} \cdot (-1)^{X_i(\mathbf{w})} \mid \mathbf{e}_{i+1} = e] \qquad \text{(Claim24)} \\
&= \frac{1}{nd} \sum_{(a,b,c) \in E} (-1)^{1[b \in S]} \cdot \mathbb{E}_{\mathbf{w}}[(-1)^{X_i(\mathbf{w})} \mid \mathbf{v}_i = a] \\
&= \frac{1}{d} \sum_{(a,b,c) \in E} x_a y_b z_c. \qquad (9)
\end{aligned}$$

Next, directly from the definition of$y$ and Claim24,

$$\mathbb{E}_{\mathbf{w}}[(-1)^{S_i(\mathbf{w})}] = \frac{1}{n} \sum_{a \in [n]} y_a. \quad (10)$$

Similarly,

$$\mathbb{E}_{\mathbf{w}}[(-1)^{X_i(\mathbf{w})}] = \frac{1}{n} \sum_{a \in [n]} x_a. \quad (11)$$

Equation(8) follows from Equations(9)–(11) and the fact that$\sum_{a \in [n]} z_a = 1$. The desired result then follows Definition12. □

5.3 Decoding C

We follow the work of Jernoimo et al. who gave a near-linear time list- and unique-decoding algorithm for Ta-Shma's code via an efficient weak regularity lemma, and prove that their algorithm also applies to our code as well. In the language ofτ-sampling distributions, they prove:[7]

Theorem35 ([20]). There exists a constantc_{JST} such that the following holds for any integersd, t, k, n and any$\tau, \varepsilon_0, \varepsilon > 0$. Let$C_0 : \mathbb{F}_2^k \to \mathbb{F}_2^n$ be a code with bias at mostε_0 which is uniquely decodable to within distance$(1 - \varepsilon_0)/4$ in time$T_0 = T_0(n, \varepsilon_0)$. Let$\mathbf{W} \sim [n]^t$ be a homogeneousτ-sampling distribution, let$C = \mathrm{dsum}_{\mathbf{W}}(C_0)$ be the corresponding direct sum lifting, and assume that the bias of$C$ is at mostε. Letβ be such that

$$\beta \geq \max\left\{\sqrt{\varepsilon}, \sqrt{c_{\mathrm{JST}} \cdot t^3 \tau}, 2\left(\frac{1}{2} + 2\varepsilon_0\right)^t\right\}.$$

Then, there exists a randomized algorithm, which given$\tilde{y} \in \mathbb{F}_2^{|\mathbf{W}|}$, recovers the list$C \cap B(\tilde{y}, 1/2 - \beta)$ with probability at least$1 - \frac{1}{\varepsilon} \cdot 2^{-\Omega(\varepsilon_0^2 n)}$ in time$\widetilde{O}(c_{\beta,t,\varepsilon_0} \cdot (|\mathbf{W}| + T_0))$, for$c_{\beta,t,\varepsilon_0} = (6/\varepsilon_0)^{2^{O(t^3/\beta^2)}}$. Moreover, if we are able to set$\beta = (1+\varepsilon)/4$, we can uniquely decode$C$ to within distance$(1-\varepsilon)/4$ with probability at least$1 - 2^{-\Omega(\varepsilon_0^2 n)}$. Plugging-in our code$C$ (using$\mathbf{W}_{H,t}$ andC_0 as described in Section5.1), we get our main result [Theorem36, stated above].

Proof of Theorem36. By Lemma34, our parity sampler$\mathbf{W}$ which we use for$C$ isτ-sampling for

$$\tau = \frac{\lambda}{4} = O\left(\frac{\log d}{\sqrt{d}}\right) = \widetilde{O}(2^{-2\sqrt{\log(1/\varepsilon)}}),$$

where we used the fact that our randomized construction of$H$ gives us$\lambda = O((1/\sqrt{d})\log d)$ and the bound on$d$ from Equation(7). All that is left is to show how the two items follow from Theorem35. For the list decoding result, we takeβ to be as small as possible. For us,

$$\left(\frac{1}{2} + 2\varepsilon_0\right)^t \leq 2^{-\frac{1}{2}\sqrt{\log(1/\varepsilon)}},$$

and$\sqrt{c_{\mathrm{JST}} \cdot t^3 \tau} = \widetilde{O}(2^{-\sqrt{\log(1/\varepsilon)}})$. We can thus conclude that$\beta \leq 2^{-\frac{1}{2}\sqrt{\log(1/\varepsilon)}}$. (Again, we are assumingε is smaller than some small constant.) For the running time, note that

$$T_0 = \exp(\mathrm{poly}(1/\varepsilon_0))n = \exp\left(2^{O(\sqrt{\log(1/\varepsilon)})}\right) \cdot k,$$

and$c_{\beta,t,\varepsilon_0}$ is thus triply-exponential in$\sqrt{\log(1/\varepsilon)}$, and overall$\widetilde{O}(c_{\beta,t,\varepsilon_0} \cdot (|\mathbf{W}| + T_0)) = \widetilde{O}(c_{\beta,t,\varepsilon_0} \cdot k)$. For the unique decoding result, we take$\beta = (1+\varepsilon)/4$, and the running time becomes doubly-exponential in$\sqrt{\log(1/\varepsilon)}$. □

[7] Jernoimo et al. prove their result under stronger requirements, however one can verify that the mixing requirement suffices.

A Splittability and τ-Sampling

Our decoding result follows from the work of Jernoimo, Srivastava, and Tulsiani[20]. In[20], for the result of Theorem35 to hold, they require$\mathbf{W}$ to beτ-splittable. τ-splittability is stronger than, and in fact implies,τ-sampling. We will not define splittability here, since for their result to hold, only a "Splittable Mixing Lemma" is required. The mixing property used in[20] goes as follows.

Definition56 (splittable mixing for ± 1 cut functions). Given positive integersn, t, and$s < t - 1$, we denote by$\mathcal{F}_s$ the set of all functions$f : [n]^t \to \{1, -1\}$ of the form

$$f(x_1, \ldots, x_t) = b \cdot \chi_{A_1}(x_1) \cdots \chi_{A_s}(x_s) \cdot \chi_B(x_{s+1}, \ldots, x_t), \tag{27}$$

where$b \in \{1, -1\}$, $A_1, \ldots, A_s \subseteq [n]$, $B \subseteq [n]^{t-s}$, and where$\chi_S(x) = -1$ if$x \in S$ and1 otherwise. For a distribution$\mathbf{W} \sim [n]^t$, we denote by$\mathbf{W}_{[i,j]}$ the distribution over$[n]^{j-i+1}$ that is obtained by sampling$\mathbf{x} \sim \mathbf{W}$ and outputting$\mathbf{x}_{[i,j]}$. For simplicity, we abbreviate$\mathbf{W}_i = \mathbf{W}_{[i,i]}$. Given$s \in [t-1]$, let$\nu_s$ be the probability measure over$[n]^t$ for which$\nu_s(x) = \prod_{i \in [s]} \Pr[\mathbf{W}_i = x_i] \cdot \Pr[\mathbf{W}_{[s+1,t]} = x_{[s+1,t]}]$. We say$\mathbf{W} \sim [n]^t$ satisfies splittable mixing with errorτ, if for every$s \in [t-1]$ and every$f, f' \in \mathcal{F}_s$ it holds that

$$\left| \mathbb{E}_{\mathbf{x} \sim \nu_s}[f(\mathbf{x}) f'(\mathbf{x})] - \mathbb{E}_{\mathbf{x} \sim \nu_{s-1}}[f(\mathbf{x}) f'(\mathbf{x})] \right| \leq \tau. \tag{28}$$

In[20], they show that the above property follows fromτ-splittablility. Here we show that the above property also follows from our weaker notion ofτ-sampling. For the sake of being compatible with[20], we will use a slightly different definition ofτ-sampling than what was given in Definition32.

Definition57. We say that a distribution$\mathbf{W}$ over$[n]^t$ isτ-sampling if for any$i \in [t-1]$, $S \subseteq [n]$, and$X \subseteq [n]^{t-i}$,

$$\mathrm{Cov}_{\mathbf{w} \sim \mathbf{W}}[1[\mathbf{w}_i \in S], 1[(\mathbf{w}_{i+1}, \ldots, \mathbf{w}_t) \in X]] \leq \tau.$$

For decoding lifted sum codes, both definitions are essentially the same, since we can always work with$\mathbf{W}_{\mathrm{rev}}$, wherein we sample$\mathbf{w} = (\mathbf{w}_1, \ldots, \mathbf{w}_t) \sim \mathbf{W}$ and output$(\mathbf{w}_t, \ldots, \mathbf{w}_1)$. Moreover, our$\mathbf{W}$ satisfies$\mathbf{W} = \mathbf{W}_{\mathrm{rev}}$.

Claim58. Let$\mathbf{W} \subseteq [n]^t$ be homogeneous andτ-sampling. Then,$\mathbf{W}$ satisfies splittable mixing with error4τ.

Proof. Let$s \in [t-1]$, $b, b' \in \{1, -1\}$ and sets$A_1, \ldots, A_s, A'_1, \ldots, A'_s \subseteq [n]$ and$B, B' \subseteq [n]^{t-s}$ that correspond to the functions$f$ andf'. For$i \in [s]$ denote$C_i = A_i \triangle A'_i$ and$D = B \triangle B'$. We then have

$$f(x) f'(x) = \sigma \cdot \chi_{C_1}(x_1) \cdots \chi_{C_s}(x_s) \cdot \chi_D(x_{s+1}, \ldots, x_t),$$

for$\sigma = bb' \in \{1, -1\}$. The subtraction on the left hand side of Equation(28)

now reads

$$\sum_{x\in[n]^{s-1}} \prod_{i\in[s-1]} \Pr[\mathbf{W}_i = x_i] \cdot \sigma \cdot \chi_{C_1}(x_1) \cdots \chi_{C_{s-1}}(x_{s-1}) \cdot \sum_{y\in[n], z\in[n]^{t-s}} \chi_{C_s}(y) \cdot \chi_D(z) \cdot (\Pr[\mathbf{W}_s = y] \Pr[\mathbf{W}_{[s+1,t]} = z] - \Pr[\mathbf{W}_{[s,t]} = y \circ z]). \quad (29)$$

The second line of the above expression can be written as

$$\mathbb{E}_{\mathbf{y}\sim\mathbf{W}_s}[(-1)^{1[\mathbf{y}\in C_s]}] \cdot \mathbb{E}_{\mathbf{z}\sim\mathbf{W}_{[s+1,t]}}[(-1)^{1[\mathbf{z}\in D]}] - \mathbb{E}_{(\mathbf{y},\mathbf{z})\sim(\mathbf{W}_s,\mathbf{W}_{[s+1,t]})}[(-1)^{1[\mathbf{y}\in C_s]}(-1)^{1[\mathbf{z}\in D]}], \quad (30)$$

which is simply the covariance between the random variables$(-1)^{1[\mathbf{y}\in C_s]}$ and$(-1)^{1[\mathbf{z}\in D]}$ where$\mathbf{y}$ and$\mathbf{z}$ are drawn appropriately. As$\mathbf{W}$ isτ-sampling, we know that

$$|\operatorname{Cov}[1[\mathbf{y} \in C_s], 1[\mathbf{z} \in D]]| \leq \tau,$$

and so Equation(30), in absolute value, amounts to

$$|\operatorname{Cov}[(-1)^{1[\mathbf{y}\in C_s]}, (-1)^{1[\mathbf{z}\in D]}]| \leq 4\tau,$$

which readily follows from the fact that$(-1)^b = 1-2b$ for$b \in \{0,1\}$, as we argued in Lemma34. Taking absolute value, by the triangle inequality, Equation(29) can be bounded by

$$4\tau \cdot \sum_{x\in[n]^{s-1}} \prod_{i\in[s-1]} \Pr[\mathbf{W}_i = x_i].$$

As$\mathbf{W}$ is homogeneous,$\Pr[\mathbf{W}_i = x_i] = 1/n$, and we are done. □

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